

Variable Selection for Gaussian Process Regression

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Summary

Variable selection is crucial in statistical modelling, as it identifies the most relevant variables for model construction. The main thrust of this study is to develop a framework that uses Gaussian Process Regression (GPR) to select significant variables. This framework involves setting spike and slab priors on the inverse length-scale parameters in the Automatic Relevance Determination (ARD) kernel, including a binary inclusion parameter to directly indicate the variables' significance, which can be identified with higher posterior inclusion probabilities. We propose a novel Metropolis independence sampler with Laplace approximation, a method not previously used within GPR to sample from the posterior distribution. In the empirical analysis, we use football data to evaluate the predictive performance and variable selection capability of our proposed framework: GPR with spike and slab priors (GPR-SS). In the simulation study, GPR-SS demonstrates higher predictive accuracy than standard GPR, as measured by Mean Squared Error (MSE). In the real application, we compare GPR-SS with Ordinary Least Squares (OLS), Linear Regression with spike and slab priors (LR-SS), and the standard GPR, finding that GPR-SS has the highest predictive accuracy and effective variable selection capability.

Keywords: Variable Selection, Gaussian Process Regression, Spike and Slab Priors, Metropolis Independence Sampler, Laplace Approximation, Empirical Analysis

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Chapter 1

Introduction

1.1 Variable Selection in Linear Regression Models

Variable selection is an important step in statistical modelling processes. It identifies the most important variables that contribute to a model's predictive power, thereby reducing dimensionality and improving interpretability.

Linear Regression is the most straightforward model for fitting data, with extensions such as Ridge Regression and the Least Absolute Shrinkage and Selection Operator (LASSO). Ridge Regression works by shrinking the coefficients of each variable towards, but not precisely zero. This retains all variables while reducing their influence, making Ridge Regression unsuitable for variable selection. In contrast, LASSO can shrink some coefficients precisely to zero, effectively performing variable selection by removing those variables.

However, both Ridge Regression and LASSO determine variable selection based on the magnitude of the coefficients, which is an indirect approach. Ideally, we would prefer a method that directly indicates whether a variable is selected. One such technique involves assigning a binary parameter γ to each variable. If γ equals 1, the variable is selected; if γ equals 0, the variable is not selected. This technique can be implemented by adding spike and slab priors to the coefficients in Linear Regression, which will be explained in detail later in this paper.

Variable selection can also treated as a specific case of the model selection problem, where each model being considered corresponds to a unique subset of the variables (Chipman et al., 2001). Model space prior represents all possible combinations of vari-

ables through a binary vector γ. Each γ corresponds to a specific subset of variables, essentially defining a candidate model. It assigns a prior $p(\gamma)$ to each possible variable subset, which can be viewed as a prior "weight". Within the Bayesian variable selection framework, the model space prior guides the computation of the posterior probability distribution, which tends to favour those variable subsets that have higher prior probabilities and better align with the observed data. Model space prior is straightforward to define, significantly reduces the computational demands, and frequently produces reasonable outcomes (Raftery et al., 1997). However, even with the implementation of model space priors, the computational complexity associated with the exponential growth in the number of variable combinations, which often leads to NP-hard problems, cannot be fully avoided (Natarajan, 1995). When the number of variables p is large, the total number of possible models is 2^p, making it impractical to compute the posterior probabilities for all possible combinations in practice. Moreover, multicollinearity is a common issue in high-dimensional datasets. In such cases, model space priors may struggle to effectively distinguish the significance of individual variables, as correlated variables tend to be either included or excluded from the model together (Miller, 2002). Model space priors often rely on stochastic search methods, which can become trapped in local optima or require substantial computational resources to thoroughly explore the space, particularly when dealing with complex model structures (George and McCulloch, 1997).

1.2 Variable Selection in Gaussian Process Regression Models

Linear models are parametric methods that impose assumptions on the data distribution and constrain the model's form. To more flexibly capture the characteristics of the data, this study employs Gaussian Process Regression (GPR) to select variables, a non-parametric approach that offers several advantages.

Firstly, the non-parametric nature makes Gaussian Processes (GPs) be able to complex, non-linear relationships within the data, making them more adaptable to various types of data compared to linear models. This flexibility enables GPs to capture intricate patterns that parametric models may overlook.

Secondly, GPs allow for the incorporation of prior knowledge, which refers to information or assumptions about the data that are known beforehand. This is achieved by selecting appropriate kernel functions that can encode various assumptions about the data, such as smoothness or periodicity. Consequently, this feature enhances the model's ability to capture and reflect the underlying structure of the data.

In summary, while linear models are straightforward and assume a specific functional form, GPs provide greater flexibility by modelling complex, non-linear relationships and integrating prior knowledge via kernel functions. This flexibility makes GPs more adaptable and effective in exploring intricate data patterns compared to linear models.

1.3 Posterior Distribution Sampling Techniques

In Bayesian statistics, the parameters' posterior distribution is proportional to the product of the likelihood function and the parameters' prior distribution. However, the posterior distribution is often intractable or has no closed form (i.e., it cannot be expressed as a known statistical distribution), making direct sampling from the posterior challenging. Variational Bayes and Markov Chain Monte Carlo (MCMC) are two main methods to address this issue.

Variational Bayes (VB) is a method that approximates the posterior distribution using a simpler, known distribution. The goal is to identify the distribution within a specified family closest to the true posterior by minimising the Kullback-Leibler divergence between the approximate and the true posterior.

Markov Chain Monte Carlo (MCMC) generates samples from an intractable posterior distribution by constructing a Markov chain with the posterior distribution as its equilibrium distribution. Common MCMC algorithms include the Metropolis-Hastings algorithm and the Gibbs sampler. In this paper, we employ an improved MCMC algorithm to sample the parameters from the posterior distribution.

The remainder of this paper is organised as follows: Chapter 2 provides an overview of existing related works. Chapter 3 explains the methodology of our proposed approach. Chapter 4 presents the results of the empirical analysis. Finally, Chapter 5 concludes the paper and provides potential future research directions.

Chapter 2

Literature Review

2.1 Spike and Slab Priors

Effective variable selection can reduce redundant information, improve predictive performance, and enhance interpretability and robustness. Several methods are available in the literature to achieve variable selection.

Firstly, filter methods select features by assessing their statistical significance with the target variable. Common techniques involve computing correlation coefficients (e.g., Pearson's correlation), conducting hypothesis tests (e.g., Chi-square test), and performing variance analysis (e.g., ANOVA) to identify the features that most significantly influence the outcome (Sánchez-Maroño et al., 2007). Secondly, wrapper methods select features by iteratively evaluating different subsets of features through model training and performance testing. The process involves statistical techniques to compare model performance metrics (e.g., accuracy, AUC) and select the subset that optimises the model's predictive ability (Mehmood et al., 2012). Finally, embedded methods combine the characteristics of the previous two methods by incorporating feature selection directly into the model training process. These methods use regularisation techniques (e.g., LASSO) that penalise less important features, thus statistically shrinking their coefficients and effectively reducing the feature set to those most relevant to the predictive model (Fonti and Belitser, 2017).

From a Bayesian perspective, Tibshirani (1996) put forward that LASSO can be interpreted as a Laplace prior on the regression coefficients. This prior is characterised by a peak at zero and heavy tails. Such a prior facilitates generating sparse solutions,

where coefficients are shrunk to zero, thereby achieving feature selection. Another variable selection approach in Bayesian statistics is spike and slab priors (Ishwaran and Rao, 2005). This prior achieves variable selection by assigning two distinct prior distributions to the parameters. The spike distribution is centred at zero and captures the sparsity of the variables. The slab distribution is flatter and identifies the variables significantly impacting the model. Compared to the Laplace prior, spike and slab priors allow for more direct variable selection by incorporating a binary inclusion parameter γ. When γ equals 1, the regression parameter is drawn from the slab distribution, indicating the inclusion of the variable. Conversely, when γ equals 0, the regression parameter is drawn from the spike distribution, which is very close to zero, indicating the exclusion of the variable.

George and McCulloch (1993) applied a Gaussian mixture prior, a type of spike and slab priors, to the linear regression coefficients. They utilised Gibbs sampling to draw samples from the full conditional posteriors of the parameters, obtaining the regression coefficient and the inclusion parameter for each variable. Even though parsimonious regression models can perform well, they often rely on certain assumptions about the data distribution and impose constraints on the model's form.

2.2 Gaussian Processes with Spike and Slab Priors

Parametric models have certain advantages in terms of interpretability. However, simple parametric models may lack the expressive power required for complex datasets. In contrast, more complex models, such as feedforward neural networks, may be challenging to implement in practice. The advent of kernel machines, such as GPs, has enhanced the flexibility of models which are practical to work with (Rasmussen, 2003).

Linkletter et al. (2006) employed GPs for variable selection by placing an independent mixture prior of a standard uniform and a point mass at one on the correlation coefficients ρ_k in the generalised exponential kernel.

$$\pi(\rho_k) = \gamma I[0 \leq \rho_k \leq 1] + (1 - \gamma)\delta_1(\rho_k) \tag{2.1}$$

However, unlike the spike and slab priors, the parameter γ associated with the standard uniform distribution is not a binary variable but rather a pre-specified prior

inclusion probability. Consequently, it does not facilitate the direct identification of variable inclusion. Andersen et al. (2014) proposed a novel prior formulation known as the structured spike and slab prior. This method allows for the integration of prior knowledge about sparsity patterns by introducing a spatial GP on the spike and slab probabilities.

$$p(x_i \mid z_i) = (1 - z_i)\delta(x_i) + z_i N\left(x_i \mid 0, \tau_0\right), \quad z_i \sim \text{Bernoulli}\left(g(\gamma_i)\right), \tag{2.2}$$

where g is a suitable injective transformation. Dance and Paige (2022) selected the automatic relevance determination (ARD) kernel as the covariance function for GPs. They applied spike and slab priors to the inverse length-scale parameters θ of the ARD kernel to achieve variable selection.

$$k_{\tau,\theta}(x, x') = \tau h(\|\theta \odot (x - x')\|_2), \tag{2.3}$$

$$p(\theta, \gamma) = \prod_{l=1}^{d} \left[\mathcal{P}_{\text{slab}}(\theta_l)\pi\right]^{\gamma_l} \left[\delta_0(\theta_l)(1 - \pi)\right]^{1-\gamma_l}, \tag{2.4}$$

where $\tau \in \mathbb{R}_{>0}$ denotes the scale of the kernel, and $h(\cdot)$ is a continuous function. Rasmussen and Williams (2006) pointed out that the inverse length-scale parameter θ determines the relevance of a variable. As θ increases, the relevance of the variable increases. When θ is estimated to be small, the variable has a tiny influence on the covariance kernel and can be effectively excluded from the model. Therefore, we adopt the ARD kernel as the covariance function and impose spike and slab priors on the inverse length-scale parameters. This approach allows the binary parameters within the spike and slab priors to facilitate the identification of variable inclusion, especially when the inverse length scales of each dimension are not distinctly different.

2.3 Computation Methods for Inference

In Bayesian statistics, the moments and percentiles of the posterior distribution are commonly used as estimates; therefore, sampling from the posterior distribution of the parameters becomes a primary task. However, direct sampling becomes challenging since the posterior distribution is often intractable and not available in closed form.

Two main methods can address this difficulty.

One approach is Variational Bayes (VB), which approximates the posterior distribution by finding the nearest distribution within a simpler family. By converting the sampling problem into an optimisation problem, VB makes computation feasible for large datasets, optimising a lower bound on the marginal likelihood (ELBO) to approximate the true posterior (Penny et al., 2006). Another approach is Markov Chain Monte Carlo (MCMC), which generates samples through a stochastic process that forms a Markov chain, eventually converging to the target posterior distribution (Geyer, 1992). Methods like the Metropolis-Hastings algorithm and Gibbs sampling are flexible and accurate.

Titsias and Lázaro-Gredilla (2011) introduced an effective variational EM algorithm for GPs with spike and slab priors in a multi-task learning framework. This algorithm identifies sparse linear combinations of GPs and thus can take advantage of the tractability of linear models. Titsias and Lawrence (2010) presented a variational inference framework for training GP models with latent variables. This study proposes a variational distribution in a factorised Gaussian form to approximate the posterior distribution of the latent variables, thereby maximising Jensen's lower bound. Compared to previous work, the method introduced by Dance and Paige (2022) runs faster than stochastic variational GPR (Hensman et al., 2013). The method is tailored to large-scale and high-dimensional datasets and uses a continuous approximation to Dirac spike and a structured variational approximation to the posterior.

Although Variational Bayes runs faster, MCMC is more accurate because it only suffers from Monte Carlo error. Gibbs sampler (Casella and George, 1992) requires all conditional distributions to be of known and easily sampled forms. However, in GPR with spike and slab priors, the posterior distribution of the parameters is complex, making Gibbs sampler infeasible in our context. Rossi et al. (2021) developed a scalable MCMC method for GPR using Hamiltonian Monte Carlo. This method requires differentiable random variables, and thus, it cannot be applied to spike and slab priors due to the presence of binary inclusion parameters, which are not differentiable. Pillai et al. (2011) designed a random walk Metropolis algorithm for GP priors, but such an approach method requires substantial time to tune the proposal distribution's standard deviation, as it significantly affects the MCMC sampling acceptance rate. Adaptive

MCMC algorithms (Roberts and Rosenthal, 2009) attempt to address this issue by automatically "learning" better parameter values of MCMC algorithms while they run. In adaptive MCMC, the proposal distribution (for a d-dimension target distribution $\pi(\cdot)$) is composed of two components.

$$Q_n(\theta, \cdot) = (1 - \beta)N(\theta, (2.38)^2\Sigma_n/d) + \beta N(\theta, (0.1)^2 I_d/d), \tag{2.5}$$

where Σ_n is the current empirical estimate of the sample covariance matrix based on the iterations so far, and β is a small positive constant. By incorporating the sample covariance information into the proposal distribution, parameter convergence can be automatically achieved with an appropriate acceptance rate.

Within the framework of GPR with spike and slab priors, we attempted to sample from the posterior distribution of the parameters using the random walk Metropolis and adaptive MCMC algorithms. However, the performance of both methods was not good; the sampling process became trapped in local optima, leading to persistent rejection of proposals and consequently reducing the sampling acceptance rate. Therefore, we propose sampling using an alternative variant of the Metropolis-Hastings algorithm, the Metropolis independence sampler. We also employ the Laplace approximation (Rue et al., 2009) to approximate the posterior distribution and use this approximated posterior as the proposal distribution. To the best of our knowledge, the independence sampler has not yet been applied to GPR with spike and slab priors in existing work. Our proposed sampling algorithm appears to be achieving faster convergence in the MCMC chains. This computation scheme provides reasonable results and exhibits good performance in terms of MCMC.

Chapter 3

Methodology

In this chapter, we will introduce our proposed framework of GPR-SS for variable selection.

3.1 Gaussian Process Regression

Consider a set $\mathcal{X}$. A Gaussian process is defined as a random function $f \sim GP(\mu(x), k_\alpha(x, x'))$, characterised by a mean function $\mu(\cdot) : \mathcal{X} \to \mathbb{R}$ and a covariance function (or 'kernel') $K(\cdot, \cdot)$. This process ensures that any finite collection of points follows a multivariate Gaussian distribution: $f = [f(x_1), \ldots, f(x_n)] \sim N(\mu, K)$, where $[\mu]_i = \mu(x_i)$ and $[K]_{ij} = k_\alpha(x_i, x_j)$, with α representing the kernel hyperparameters. The covariance kernel is a positive definite function $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, i.e.,

$$\sum_{i=1}^{n} \sum_{j=1}^{n} K(x_i, x_j) a_i a_j \geq 0 \tag{3.1}$$

for all $x_1, \ldots, x_n \in \mathcal{X}$ and all scalars $a_1, \ldots, a_n$.

Given a dataset $\mathcal{D} = (y, X) \in \mathbb{R}^n \times \mathbb{R}^{n \times d}$, a Gaussian process model is defined as follows:

$$y = f(x) + \epsilon, \quad \epsilon \sim N(0, \sigma^2 I_n), \tag{3.2}$$

$$f \mid \alpha \sim N(\mu, k_\alpha(x, x')) \tag{3.3}$$

If the response vector $y = (y_1, \ldots, y_n)$ is assumed to be centred and scaled, such that

it has a mean of 0 and a variance of 1, then the mean μ is set to 0.

In standard GPR, the observation model is a Gaussian distribution centred at f: $p(y \mid f) = \prod_{i=1}^{n} N(y_i \mid f_i, \sigma^2)$. In this context, the marginal likelihood possesses a closed form $p_{\alpha,\sigma}(y) = N(y \mid \mu, K + \sigma^2 I_n)$ and can be maximised to learn the kernel parameters α and the noise variance σ^2 (referred to as ML-II optimisation)(Rasmussen and Williams, 2006).

3.2 Automatic Relevance Determination Kernel

Automatic Relevance Determination (ARD) kernels are commonly employed to perform variable selection in GPR (MacKay et al., 1998). Most ARD kernels, such as the squared exponential (SE), Matérn, and Cauchy kernels, belong to a subclass of anisotropic stationary kernels of the form (Dance and Paige, 2022):

$$k_{\tau,\theta}(x, x') = \tau h(\|\theta \odot (x - x')\|_2). \tag{3.4}$$

Here, $\theta \in \mathbb{R}_+^d$ represents per-dimension inverse length scales, which serve as a measure of variable relevance. The parameter $\tau \in \mathbb{R}_{>0}$ denotes the scale of the kernel, and $h(\cdot)$ is a continuous (monotonically decreasing) function.

In our proposed GPR framework, we choose the ARD kernel to be in the form:

$$k_{\rho,\lambda}(x, x') = \rho^2 \exp\left(-\frac{1}{2}\sum_{l=1}^{d} \lambda_l^2 (x_l - x_l')^2\right) \tag{3.5}$$

In the "sklearn" package of Python, we use the RBF() command to represent the ARD kernel, which by default sets the value of ρ to 1; hence, we explicitly set $\rho = 1$ in our model.

In ML-II optimisation, irrelevant dimensions typically result in $\lambda_l \to 0^+$ due to the intrinsic complexity penalty of the marginal likelihood (Rasmussen and Williams, 2006). Consequently, variable selection can be achieved by hard-thresholding $\{\lambda_l\}_{l=1}^d$. However, determining an appropriate threshold in high-dimensional settings is challenging and may significantly reduce predictive accuracy if not done reliably (Dance and Paige, 2022). Additionally, when the differences between λ values are not substantial, it becomes difficult to ascertain variable selection directly. Therefore, a more

direct method for variable selection is necessary.

Dance and Paige (2022) observed that the alternative variable-relevance measures, as Paananen et al. (2019) developed, utilise posterior predictive sensitivity. However, these methods are constrained by the necessity of determining an appropriate threshold for binary selection decisions.

3.3 Spike and Slab Priors

Spike and slab priors, as introduced by Mitchell and Beauchamp (1988), are widely regarded as the gold standard for high-dimensional variable selection problems due to their remarkable properties (Narisetty and He, 2014; Castillo et al., 2015). These priors offer superior empirical performance and interpretability, making them a preferred choice in such contexts.

Spike and slab priors are a type of hierarchical Bayesian prior used extensively in high-dimensional variable selection. The 'spike' component represents a concentrated mass at zero, encouraging sparsity by shrinking coefficients of irrelevant predictors towards zero. Conversely, the 'slab' component is a diffuse, continuous distribution that allows for the inclusion of significant predictors with non-zero coefficients. This dual-component structure effectively balances the need for both sparsity and flexibility, enabling the model to distinguish between relevant and irrelevant variables while maintaining interpretability and robustness.

For a likelihood $p(y \mid \theta)$ with per-dimension 'relevance' parameters $\theta \in \Theta \subseteq \mathbb{R}^d$, spike and slab priors enhance the model by introducing binary inclusion variables $\gamma \in \{0,1\}^d$. When $\gamma = 1$, it indicates that the variable is included in the model, whereas $\gamma = 0$ signifies that the variable is excluded. The prior distribution $p(\theta, \gamma)$ is then defined as follows:

$$p(\theta, \gamma) = \prod_{l=1}^{d} \left[\mathcal{P}_{\text{slab}}(\theta_l)\pi\right]^{\gamma_l} \left[\delta_0(\theta_l)(1-\pi)\right]^{1-\gamma_l} \tag{3.6}$$

In this formulation, $\mathcal{P}_{\text{slab}}(\theta_l)$ denotes the slab component, $\delta_0(\theta_l)$ represents the spike component, and π is the prior inclusion probability (Dance and Paige, 2022). This structure effectively combines the strengths of both components to manage sparsity and retain significant predictors.

For the per-dimension inverse length-scale parameter λ_l in our ARD kernel, we assign a prior where both the spike and slab components follow a normal distribution. Since the inverse length-scale parameters λ are positive, we use truncated normal distributions in our priors.

$$\lambda_l \mid \gamma_l \sim \gamma_l N_{>0}(0, \psi^2) + (1 - \gamma_l) N_{>0}(0, \omega^2), \quad \omega << \psi, \tag{3.7}$$

where each λ_l is associated with a γ_l. The variance of the 'spike' component is much smaller than that of the 'slab' component. Therefore, when γ_l equals 0, the λ_l is drawn from the 'spike' component, which is very close to the mean 0. When gamma equals 1, the λ_l is drawn from the 'slab' component, resulting in a larger value. This behaviour is also consistent with the characteristics of the ARD kernel.

3.4 Posterior Computation

Denote ϕ as a random variable, which is assigned a prior probability density function $\pi(\phi)$. The prior distribution reflects the beliefs about ϕ before any data is observed.

The posterior distribution represents the updated beliefs about ϕ after the data has been observed and serves as the main focus for inference. It is mathematically expressed as:

$$\pi(\phi \mid y) = \frac{f(y \mid \phi)\pi(\phi)}{f(y)}, \tag{3.8}$$

where $f(y) = \int_\phi f(y \mid \phi)\pi(\phi)d\phi$.

The term $f(y)$ is known as the marginal likelihood or evidence and indicates the probability of the observed data under the chosen probabilistic model. This term plays a crucial role by acting as a normalising constant, ensuring that the posterior distribution is a valid probability measure. Since $f(y)$ is a constant with respect to ϕ, the posterior distribution is proportional to the product of the prior distribution and the likelihood.

$$\pi(\phi \mid y) \propto f(y \mid \phi)\pi(\phi) \tag{3.9}$$

Given the full model:

$$y = f(x) + \epsilon, \quad \epsilon \sim N(0, \sigma^2 I_n), \tag{3.10}$$

$$f = f(x) \sim N(0, K(x, x')), \tag{3.11}$$

$$K(x, x') = \exp\left(-\frac{1}{2}\sum_{l=1}^{d} \lambda_l^2 (x_l - x_l')^2\right) \tag{3.12}$$

In our experiments, we fix the value of σ^2 to ensure model simplicity and stability. Fixing σ^2 reduces the number of hyperparameters, simplifying the model and making the training process more stable, particularly in noisy data. This approach also helps prevent overfitting, as it imposes a regularisation effect by constraining the model's capacity to fit the noise in the training data.

We assign the spike and slab priors for each λ_l:

$$\lambda_l \sim \gamma_l N_{>0}(0, \psi^2) + (1 - \gamma_l) N_{>0}(0, r\psi^2), \quad l = 1, \ldots, d, \tag{3.13}$$

where $r = 10^{-4}$, ψ^2 is a positive known constant. This formulation deviates from the standard spike and slab priors, which typically employ normal distributions. Here, we introduce half-normal distributions to accommodate the positivity constraints on λ. The variable inclusion indicators $\gamma = (\gamma_1, \ldots, \gamma_d)$ are independent and distributed as

$$\gamma_l \sim \text{Bernoulli}(\pi_\gamma), \quad l = 1, \ldots, d, \tag{3.14}$$

where π_γ is a user specified probability. There is a class of hyper g-priors that assign a Beta distribution to π_γ, which can be considered here. However, the Bernoulli prior on γ with a pre-specified π_γ is mainly used for simplicity.

In GPs, the joint distribution of f and y follows a multivariate normal distribution:

$$\begin{pmatrix} f \\ y \end{pmatrix} \sim N\left(\begin{pmatrix} \mathbf{0}_n \\ \mathbf{0}_n \end{pmatrix}, \begin{pmatrix} K & K \\ K & K + \sigma^2 I_n \end{pmatrix}\right). \tag{3.15}$$

By the property of the multivariate normal distribution, we can derive the conditional distribution of y given f:

$$y|f \sim N(f, \sigma^2 I_n). \tag{3.16}$$

When calculating the full conditional posterior of f, we only need to consider the terms related to f. Thus, the full conditional posterior of f can be obtained as follows:

$$\begin{aligned}
\pi(f|y,\lambda,\gamma) &\propto f(y|f)\pi(f|\lambda) \\
&= N(f,\sigma^2 I_n)N(0,K) \\
&\propto (\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{(y-f)^T(y-f)}{2\sigma^2}\right) \det(K)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}f^T K^{-1} f\right) \\
&\propto \exp\left(-\frac{(y-f)^T(y-f)}{2\sigma^2}\right) \exp\left(-\frac{1}{2}f^T K^{-1} f\right) \qquad (3.17)
\end{aligned}$$

For λ note that, given γ, we can write $\lambda \mid \gamma \sim N(0, D)$, where D is a $d \times d$ diagonal matrix with entries equal to ψ^2 if $\gamma_l = 1$ and $r\psi^2$ if $\gamma_l = 0$. Similarly, the full conditional posterior of λ can be obtained as follows:

$$\begin{aligned}
\pi(\lambda|y,f,\gamma) &\propto f(y|f)\pi(f|\lambda)\pi(\lambda|\gamma) \\
&= N(f,\sigma^2 I_n)N(0,K)\pi(\lambda|\gamma) \\
&\propto (\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{(y-f)^T(y-f)}{2\sigma^2}\right) \det(K)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}f^T K^{-1} f\right) \\
&\times \exp\left(-\frac{1}{2}\lambda^T D^{-1}\lambda\right) \\
&\propto \det(K)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}f^T K^{-1} f\right) \exp\left(-\frac{1}{2}\lambda^T D^{-1}\lambda\right), \qquad (3.18)
\end{aligned}$$

For the full conditional posterior of each γ_l, note that

$$\pi(\gamma_l = 1|y,\lambda,f) \propto \pi_\gamma N_{>0}(\lambda_l|0,\psi^2), \qquad (3.19)$$

$$\pi(\gamma_l = 0|y,\lambda,f) \propto (1-\pi_\gamma) N_{>0}(\lambda_l|0,r\psi^2). \qquad (3.20)$$

Hence,

$$\begin{aligned}
\pi(\gamma_l = 1|y,\lambda,f) &= \frac{\pi_\gamma N_{>0}(\lambda_l|0,\psi^2)}{\pi_\gamma N_{>0}(\lambda_l|0,\psi^2) + (1-\pi_\gamma) N_{>0}(\lambda_l|0,r\psi^2)} \\
&= \frac{1}{1 + \frac{1-\pi_\gamma}{\pi_\gamma} N_{>0}(\lambda_l|0,r\psi^2)/N_{>0}(\lambda_l|0,\psi^2)}. \qquad (3.21)
\end{aligned}$$

The posterior inclusion probability for each variable is the mean of γ_l. The threshold value of 0.5 is a suggestion and can alternatively be viewed as a tuning parameter.

To achieve the goal of prediction, we only need to sample λ and then input the sampled λ into the RBF() function in the sklearn package in Python. This method simplifies the process by focusing only on λ. Additionally, as shown in equations 3.17 and 3.18, f and λ are mutually influential, meaning that poor sampling results for f can affect the sampling of λ and vice versa. For instance, when f becomes trapped in a local optimum, it can place λ in a similar situation, reducing acceptance rates for both. Therefore, we use the marginal distribution of y instead of the conditional distribution of y given f so that there is no need to sample f, thereby improving the efficiency and robustness of the sampling process. The updated posterior distribution of λ is:

$$\begin{aligned}\pi(\lambda|y,\gamma) &\propto f(y)\pi(\lambda|\gamma) \\ &= N(0, K+\sigma^2 I_n)\pi(\lambda|\gamma) \\ &\propto \det(K+\sigma^2 I_n)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}y^T(K+\sigma^2 I_n)^{-1}y\right)\exp\left(-\frac{1}{2}\lambda^T D^{-1}\lambda\right).\end{aligned} \tag{3.22}$$

The next step involves sampling from the posterior distribution of λ. We employ the independence sampler from the Metropolis-Hastings algorithm, which will be introduced in the following section.

3.5 Improved Metropolis Independence Sampler

3.5.1 Metropolis Independence Sampler

The Metropolis-Hastings algorithm is an MCMC method to generate samples from our target posterior distribution $\pi(\lambda \mid y, \gamma)$, denoted by $\pi(\lambda \mid y)$ hereafter. The algorithm proceeds as follows:

1. **Initialisation**: Begin by setting the initial value λ_0 at $t = 0$. The choice of λ_0 can be arbitrary, but a good starting point may enhance the convergence speed.

2. **Iteration**: For each iteration t from 0 to $T-1$, perform the following steps:

   - **Proposal sampling**: Generate a candidate sample λ^* from a proposal distribution $q(\lambda^* \mid \lambda_t)$. The proposal distribution q should be chosen such

that it is easy to sample from. It can depend on the current state λ_t to propose the next state λ^*.

- **Acceptance probability**: Calculate the acceptance probability $\alpha(\lambda_t, \lambda^*)$, which determines whether the proposed sample λ^* should be accepted as the next state. The acceptance probability is given by:

$$\alpha(\lambda_t, \lambda^*) = \min\left(1,\ \frac{\pi(\lambda^* \mid y)q(\lambda_t \mid \lambda^*)}{\pi(\lambda_t \mid y)q(\lambda^* \mid \lambda_t)}\right). \tag{3.23}$$

  This formula makes the Markov chain satisfy a detailed balance, thus converging to the target posterior distribution. The ratio $\frac{\pi(\lambda^*|y)}{\pi(\lambda_t|y)}$ measures how well the proposed state λ^* fits the data compared to the current state λ_t, while the ratio $\frac{q(\lambda_t|\lambda^*)}{q(\lambda^*|\lambda_t)}$ accounts for the asymmetry in the proposal distribution.

- **Acceptance or rejection**: Accept the candidate sample with probability $\alpha(\lambda_t, \lambda^*)$ by setting $\lambda_{t+1} = \lambda^*$. If the candidate sample is not accepted, retain the current state by setting $\lambda_{t+1} = \lambda_t$. This step keeps the Markov chain ergodic, enabling it to explore the entire state space.

By iterating this process, the algorithm constructs a Markov chain whose stationary distribution is the target posterior distribution $\pi(\lambda \mid y)$. Over a sufficient number of iterations, the samples produced by the chain will approximate the target distribution.

By setting $q(\lambda^* \mid \lambda_t) = q(\lambda^*)$, we obtain the Metropolis independence sampler with acceptance probability:

$$\alpha(\lambda_t, \lambda^*) = \min\left(1,\ \frac{\pi(\lambda^* \mid y)q(\lambda_t)}{\pi(\lambda_t \mid y)q(\lambda^*)}\right) = \min\left(1,\ \frac{f(y \mid \lambda^*)\pi(\lambda^*)q(\lambda_t)}{f(y \mid \lambda_t)\pi(\lambda_t)q(\lambda^*)}\right). \tag{3.24}$$

Then, the Metropolis independence sampler can be described as follows:

3.5.2 Laplace Approximation

Let $\pi^*(\lambda \mid y) = f(y)\pi(\lambda \mid \gamma)$ represent the unnormalised posterior distribution and denote λ_M as the posterior mode. In Bayesian inference, the posterior distribution $\pi(\lambda \mid y)$ represents the probability distribution of the parameter λ given the observed data y. The posterior mode is the parameter value λ that maximises the posterior

Algorithm 1 Metropolis Independence Sampler

1: **Initialise** λ_0 at $t = 0$.
2: **for** $t = 0$ to $T - 1$ **do**
3: Sample a point λ^* from $q(\lambda^*)$.
4: Set $\lambda_{t+1} = \lambda^*$ with probability $\alpha(\lambda_t, \lambda^*)$:

$$\alpha(\lambda_t, \lambda^*) = \min\left(1,\ \frac{f(y \mid \lambda^*)\pi(\lambda^*)q(\lambda_t)}{f(y \mid \lambda_t)\pi(\lambda_t)q(\lambda^*)}\right).$$

5: Otherwise, set $\lambda_{t+1} = \lambda_t$.
6: **end for**

distribution $\pi(\lambda \mid y)$, which is given by:

$$\lambda_M = \arg\max_{\lambda} \pi^*(\lambda \mid y) \tag{3.25}$$

In other words, the posterior mode is the most likely parameter value in the parameter space. It is the peak position of the posterior distribution. Under regular conditions, λ_M satisfies the equation:

$$-\nabla_\lambda \log \pi^*(\lambda_M \mid y) = 0. \tag{3.26}$$

This implies that the gradient of the log-posterior at λ_M is zero, indicating a local maximum.

Next, let $H(\lambda)$ denote the Hessian matrix, which is the second partial derivative of the negative log-posterior. Specifically,

$$H(\lambda) = -\nabla^2_\lambda \log \pi^*(\lambda \mid y) \tag{3.27}$$

As the sample size n becomes large, the posterior distribution $\pi(\lambda \mid y)$ approaches a normal distribution centred at λ_M with a covariance matrix given by the inverse of the Hessian matrix at the mode:

$$\pi(\lambda \mid y) \to N\left(\lambda_M, H^{-1}(\lambda_M)\right) \tag{3.28}$$

This result follows from the Bayesian Central Limit Theorem, which asserts that, under regularity conditions, the posterior distribution tends towards normality as the sample

size increases.

In our study, we assume that λ are independent. The covariance of λ can be computed using numerical differentiation, and note that both analytical and automatic differentiation methods are viable alternatives.

In the Metropolis independence sampler, the choice of the proposal distribution significantly impacts the convergence efficiency and the sampling acceptance rate. Therefore, we set $N(\lambda_M,\ \sigma^2 H_*^{-1}(\lambda_M))$ (can be considered as an approximate Laplace approximation of the posterior of λ) as the proposal distribution, where H_* is the covariance matrix of λ. This proposal also maintains an acceptance rate of around 50%.

Algorithm 2 Complete Sampling Algorithm

Input: Data: (y, X), initialisation: $\boldsymbol{\lambda}$ at $t = 0$, other parameters: $\boldsymbol{\pi}_\gamma$, $\boldsymbol{\psi}$, r, where $\boldsymbol{\psi}$ is a vector

for $t = 0$ to $T - 1$ **do**
  ***Sampling process of $\boldsymbol{\gamma}$**:
  Compute $p_{\text{spike}} = N(\boldsymbol{\lambda}_t,\ r\boldsymbol{\psi})$
  Compute $p_{\text{slab}} = N(\boldsymbol{\lambda}_t,\ \boldsymbol{\psi})$
  Obtain $L_{\text{ratio}} = p_{\text{spike}}/p_{\text{slab}}$
  Sample $\boldsymbol{\gamma}$ from $Binomial(1,\ ps)$ with $ps = (1 + ((1 - \pi_\gamma) \cdot L_{\text{ratio}}/\pi_\gamma))^{-1}$
  ***Metropolis independence sampling for $\boldsymbol{\lambda}$**:
  Sample $\boldsymbol{\lambda}^*$ from $N(\boldsymbol{\lambda}_M,\ \sigma^2 H_*^{-1}(\boldsymbol{\lambda}_M))$
  while $\boldsymbol{\lambda}^* \leq 0$ **do**
    Resample $\boldsymbol{\lambda}^*$ from $N(\boldsymbol{\lambda}_M,\ \sigma^2 H_*^{-1}(\boldsymbol{\lambda}_M))$
  end while
  Compute $p_{\text{proposal}} = \log \pi(\boldsymbol{\lambda}^* \mid y)$
  Compute $p_{\text{current}} = \log \pi(\boldsymbol{\lambda}_t \mid y)$
  Generate u from a uniform distribution $U(0, 1)$
  if $p_{\text{proposal}} - p_{\text{current}} > 0$ **or** $u < \exp(p_{\text{proposal}} - p_{\text{current}})$ **then**
    Set $\boldsymbol{\lambda}_{t+1} = \boldsymbol{\lambda}^*$
  else
    Set $\boldsymbol{\lambda}_{t+1} = \boldsymbol{\lambda}_t$
  end if
  if $\boldsymbol{\lambda}^*$ is accepted **then**
    Recompute the value of kernel K_{new}:

$$K_{new}(x, x') = \exp\left(-\frac{1}{2}\sum_{l=1}^{d} \boldsymbol{\lambda}_l^{*2}(x_l - x'_l)^2\right)$$

  end if
end for

Output: $\boldsymbol{\lambda}, \boldsymbol{\gamma}$

Chapter 4

Empirical Analysis

The empirical analysis section includes the simulation study and the real application. The simulation study validates the reliability of our proposed model, while the real application demonstrates its performance on real data.

4.1 Dataset Overview

This study focuses on the offensive and passing data of forwards in the top five European leagues for the 2021-2022 season. The dataset consists of players' match data and transfer market values, which are publicly available data collected from football websites.

Players' match data are obtained from ***whoscored.com***, one of the most authoritative football data sites. Due to the complexity of retrieval, it is impractical to manually record the data item by item. Therefore, a Python crawler is used to grab the players' match data from the website. Players' transfer market values (in millions) are obtained from ***transfermarkt.com***. These values are then matched with the players' match data to compile a final dataset containing 328 observations, 14 explanatory variables, and the target variable "value" (transfer market values). The information for these 14 explanatory variables can be found in Table 4.6.

We have included the original 14 variables and their quadratic terms to expand the dataset. All 28 explanatory variables and the target variable have been standardised.

Table 4.1: *Explanatory Variables in Football Data*

Variable	Explanation	Variable	Explanation
age	Age in 2022	**goals**	Total goals
assists	Total assists	**fouls**	Fouled per game
mins	Minutes played	**shots**	Shots per game
offside	Offsides per game	**keyPass**	Key passes per game
dribbleWon	Dribbles won per game	**totalPass**	Passes per game
accurateCross	Cross success percentage	**accurateLongPass**	Long pass success percentage
turnover	Turnover per game	**accurateThrough**	Pass success percentage

4.2 Simulation Study

4.2.1 Simulation Procedure

In the simulation study, we compare the predictive outcomes of two models using Mean Squared Error (MSE) as the metric: the standard GPR (ARD kernel) model and the GPR (ARD kernel) model with spike and slab priors (GPR-SS). the variable selection performance of GPR-SS is also evaluated.

The simulation study is conducted as follows:

- We fit a linear regression model to the 328 football data observations and select significant variables with p-values less than 0.05. These four variables have also been proven to be significant in the GPR models in the real application section.
- We retain the regression coefficients for these four variables and the variance from the model, then generate new target variable values by adding six non-significant variables, which are all zeros ($\mathbf{0}_1$, $\mathbf{0}_2$, $\mathbf{0}_3$, $\mathbf{0}_4$, $\mathbf{0}_5$ and $\mathbf{0}_6$).
- We carry out 25 experiments, each involving randomly dividing the dataset into a training set and a test set using different random seeds, with a split ratio of 75% and 25%, respectively.
- For each experiment, we fit the standard GPR and GPR-SS models on the training set. Predictions are then made on the test set, and the MSE is calculated to

compare the prediction performance. The MSE is defined as:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2, \tag{4.1}$$

where y_i represents the actual values, $\hat{y}_i$ represents the predicted values, and n is the number of observations.

4.2.2 Simulation Results

Table 4.2: *MSE of 25 Experiments*

Experiment	**GPR-SS** (10^{-3})	**GPR** (10^{-3})	**Condition**
1	5.19	5.34	1
2	4.53	4.68	1
3	3.86	4.02	1
4	4.21	4.14	0
5	3.97	4.09	1
6	4.12	4.26	1
7	3.54	3.68	1
8	3.89	4.13	1
9	4.36	4.26	0
10	4.22	4.37	1
11	3.69	3.77	1
12	4.17	4.31	1
13	4.22	4.35	1
14	3.67	3.63	0
15	3.82	3.94	1
16	4.24	4.16	0
17	3.62	3.81	1
18	4.31	4.39	1
19	4.52	4.63	1
20	4.25	4.14	0
21	2.97	3.12	1
22	4.52	4.65	1
23	4.33	4.43	1
24	4.19	4.29	1
25	3.96	4.15	1
Average	4.09	4.19	0.8

In each experiment, there are 500 iterations, with an acceptance rate of around 50%. The posterior mean of λ is used as the estimate for the regression coefficients in GPR-SS. Table 4.2 presents the MSE values from 25 experiments. The 'condition' column indicates the comparative results between the two models. If GPR-SS has a lower MSE, the condition is assigned a value of 1; otherwise, it is assigned a value of 0. Out of the 25 experiments, GPR-SS exhibits a lower MSE 20 times, indicating an 80% probability that GPR-SS demonstrates superior predictive performance. The mean MSE values across these 25 experiments have also been calculated, showing that GPR-SS generally outperforms the standard GPR in the simulation study.

GPR-SS can also estimate the inclusion probability for each variable. Across the 25 experiments, it correctly identified the four significant variables: **goals**, **totalPass**, $\mathbf{age}^2$ and $\mathbf{shots}^2$ 22 times. This indicates that our model has an accuracy of over 85% in correctly identifying significant variables. In the remaining three experiments, the posterior inclusion probabilities for each variable are displayed in Table 4.3.

Table 4.3: *Posterior Inclusion Probabilities in Misidentified Experiments*

goals	**totalPass**	$\mathbf{age}^2$	$\mathbf{shots}^2$	$\mathbf{0}_1$	$\mathbf{0}_2$	$\mathbf{0}_3$	$\mathbf{0}_4$	$\mathbf{0}_5$	$\mathbf{0}_6$
0.914	0.7	**0.48**	0.88	0.14	0.12	0.02	0.14	0.05	0.008
0.94	0.64	**0.42**	0.9	0.12	0.12	0.06	0.08	0.04	0.012
0.92	0.68	**0.46**	0.86	0.08	0.134	0.1	0.1	0.08	0.01

In determining whether a variable is included, we set a threshold of 0.5. If the posterior inclusion probability exceeds 0.5, the variable is considered significant. We observe that the posterior inclusion probabilities for $\mathbf{age}^2$ are 0.48, 0.42, and 0.46, all of which are less than 0.5, but the differences are marginal. So it is ambiguous to determine the significance of variable $\mathbf{age}^2$. In the other 22 experiments where the variable selection is correctly performed, the posterior inclusion probability of $\mathbf{age}^2$ ranges between 0.5 and 0.6. One solution is to calculate the mean posterior inclusion probability of $\mathbf{age}^2$ over 25 experiments, which is 0.54. We can also set a higher threshold, such as 0.6, to more clearly select variables. The threshold value of 0.5 is just a suggestion, and an alternative is to view this as a tuning parameter.

Overall, GPR-SS demonstrates its reliability in the simulation study: it correctly selects variables with an 88% probability and achieves a lower prediction error with an 80% probability compared to the standard GPR. Next, we will apply the full football

dataset to discuss the performance of GPR-SS.

4.3 Real Application

In the real application, we compare GPR-SS with Ordinary Least Squares (OLS), Linear Regression with spike and slab priors (LR-SS), and the standard GPR, evaluating the model performance by MSE.

4.3.1 Baseline Method

LR-SS places the prior on the regression coefficients β, that is:

$$y \sim N\left|(X\beta, \sigma_\epsilon^2 I_n\right), \tag{4.2}$$

$$\beta_l \sim \gamma_l N(0, \psi^2) + (1-\gamma_l)N(0, \omega^2), \quad \omega << \psi \tag{4.3}$$

$$\gamma_l \sim Bernoulli(\pi_\gamma). \tag{4.4}$$

The other parameter settings are similar to those used in GPR-SS.

The objective is to draw samples from the posterior distribution of the parameters. In this approach, the posterior distribution has a closed form, which permits the use of the Gibbs sampler. σ_ϵ^2 is no longer fixed, and we assign the improper prior for σ_ϵ^2 such that $\pi(\sigma_\epsilon^2) \propto 1/\sigma_\epsilon^2$. Then, we apply the Gibbs sampler to sample σ_ϵ^2, β, and γ from their full conditional posteriors, which are shown as follows:

- $\gamma_l \sim \text{Bernoulli}(p_l)$, where $p_l = \left(1 + \frac{(1-\pi_\gamma)p_{\text{spike}}^{(l)}}{\pi_\gamma p_{\text{slab}}^{(l)}}\right)^{-1}, \quad l = 1, \ldots, d.$
- $\beta \sim N(b, B)$, where $B^{-1} = \frac{1}{\sigma_\epsilon^2}X^TX + D^{-1}$, $b = BX^Ty/\sigma_\epsilon^2$ and D is a $d \times d$ diagonal matrix with entries equal to ψ^2 if $\gamma_l = 1$ and $r\psi^2$ if $\gamma_l = 0$.
- $\sigma_\epsilon^2 \sim \text{IGamma}\left((n-1)/2, S\right)$, where n is the number of observations and $S = \frac{1}{2}(y - X\beta)^T(y - X\beta)$.

(Note: Given the standardisation of all explanatory variables and the target variable, the intercept term is considered unnecessary during the process.)

For the full conditional of each γ, it is the same as that of GPR-SS. The full conditional posterior of σ_ϵ^2 can be obtained by:

$$\begin{aligned}\pi(\sigma_\epsilon^2 \mid y, X, \beta, \gamma) &\propto f(y \mid \sigma_\epsilon^2, \beta, \gamma)\pi(\sigma_\epsilon^2) \\ &\propto (\sigma_\epsilon^2)^{-\frac{n}{2}} \exp\left(-\frac{(y - X\beta)^T(y - X\beta)}{2\sigma_\epsilon^2}\right)(\sigma_\epsilon^2)^{-1} \\ &\propto (\sigma_\epsilon^2)^{-\frac{n}{2}-1} \exp\left(-\frac{(y - X\beta)^T(y - X\beta)}{2\sigma_\epsilon^2}\right). \qquad (4.5)\end{aligned}$$

The above can be recognised as the inverse Gamma distribution with parameters $\frac{n}{2}$ and $\frac{1}{2}(y - X\beta)^T(y - X\beta)$.

Similar to GPR-SS, we can write $\beta \mid \gamma \sim N(0, D)$. Hence, the full conditional posterior of β is:

$$\begin{aligned}\pi(\beta \mid y, X, \gamma, \sigma_\epsilon^2) &\propto f(y \mid \sigma_\epsilon^2, \beta, \gamma)\pi(\beta \mid \gamma) \\ &\propto \exp\left(-\frac{(y - X\beta)^T(y - X\beta)}{2\sigma_\epsilon^2}\right)\exp\left(-\frac{1}{2}\beta^T D^{-1}\beta\right)\end{aligned}$$

The posterior above is a multinormal distribution with mean b and covariance matrix B, where $b = BX^T y/\sigma_\epsilon^2$ and $B^{-1} = \frac{1}{\sigma_\epsilon^2}X^T X + D^{-1}$.

4.3.2 Real Application Results

We use all 28 variables and fit four models: OLS, LR-SS, standard GPR, and GPR-SS. We perform 4-fold cross-validation and calculate the mean prediction MSE on the test set.

Table 4.4: *Mean MSE of 4-fold Cross-Validation*

Model	**Mean MSE**
OLS	0.5811
LR-SS	0.5021
GPR (ARD kernel)	0.4937
GPR-SS (ARD kernel)	0.4772

Table 4.5: *Posterior Summaries of the Regression Coefficients*

	Mean	**Median**	**95% Percentile**
λ_1	5.108×10^{-3}	5.296×10^{-3}	8.477×10^{-3}
λ_2	0.249	0.249	0.272
λ_3	0.128	0.127	0.15
λ_4	2.391×10^{-10}	2.222×10^{-10}	5.852×10^{-10}
λ_5	3.005×10^{-6}	2.668×10^{-6}	6.591×10^{-6}
λ_6	2.703×10^{-5}	2.539×10^{-5}	5.422×10^{-5}
λ_7	2.248×10^{-4}	1.914×10^{-4}	5.21×10^{-4}
λ_8	2.932×10^{-6}	2.944×10^{-6}	5.979×10^{-6}
λ_9	0.132	0.132	0.152
λ_{10}	0.233	0.232	0.261
λ_{11}	3.356×10^{-10}	3.01×10^{-10}	7.687×10^{-10}
λ_{12}	0.062	0.066	0.099
λ_{13}	2.238×10^{-6}	1.874×10^{-6}	5.522×10^{-6}
λ_{14}	0.312	0.31	0.359
λ_{15}	0.445	0.442	0.482
λ_{16}	2.627×10^{-9}	2.471×10^{-9}	5.572×10^{-9}
λ_{17}	2.686×10^{-8}	2.659×10^{-8}	5.457×10^{-8}
λ_{18}	4.951×10^{-7}	4.847×10^{-7}	8.779×10^{-7}
λ_{19}	0.161	0.161	0.183
λ_{20}	0.271	0.272	0.295
λ_{21}	2.616×10^{-9}	2.452×10^{-9}	5.546×10^{-9}
λ_{22}	3.385×10^{-8}	3.362×10^{-8}	6.612×10^{-8}
λ_{23}	2.59×10^{-6}	2.349×10^{-6}	5.511×10^{-6}
λ_{24}	4.008×10^{-6}	4.079×10^{-6}	7.206×10^{-6}
λ_{25}	2.204×10^{-10}	1.749×10^{-10}	5.845×10^{-10}
λ_{26}	0.242	0.242	0.277
λ_{27}	0.13	0.13	0.156
λ_{28}	3.151×10^{-6}	2.801×10^{-6}	6.653×10^{-6}

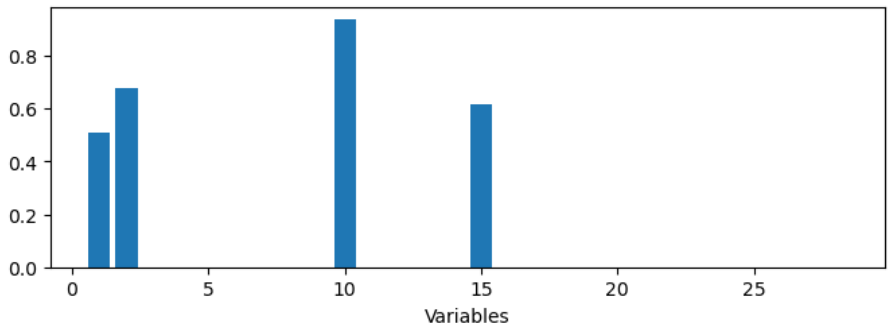


Figure 4.1: *Mean Inclusion Probabilities of LR-SS*

Table 4.6: *Mean Inclusion Probabilities of GPR-SS*

Variable	**Probability**	**Variable**	**Probability**
age	0.15	**age**^{2}	**0.932**
goals	**0.88**	**goals**^{2}	0.008
assists	**0.82**	**assists**^{2}	0.01
fouls	0.006	**fouls**^{2}	0.01
mins	0.09	**mins**^{2}	**0.85**
shots	0.12	**shots**^{2}	**0.92**
offside	0.23	**offside**^{2}	0.008
keyPass	0.08	**keyPass**^{2}	0.015
dribbleWon	**0.82**	**dribbleWon**^{2}	0.016
totalPass	**0.875**	**totalPass**^{2}	0.095
accurateCross	0.006	**accurateCross**^{2}	0.005
accurateLongPass	**0.645**	**accurateLongPass**^{2}	**0.876**
turnover	0.08	**turnover**^{2}	**0.82**
accurateThrough	**0.925**	**accurateThrough**^{2}	0.088

Table 4.5 provides the information of the estimated regression coefficients obtained from one cross-validation process. Similar to the simulation study, the posterior mean of λ is used as the estimate for the regression coefficients in GPR-SS. Each process of the cross-validation consistently identifies the same significant variables. Table 4.6 presents the mean posterior inclusion probabilities for the 28 variables in 4-fold cross-validation. Among these, 11 variables have inclusion probabilities greater than 0.5, indicating that GPR-SS has identified 11 variables that significantly influence transfer market value, effectively reducing the dimensionality by over 50%.

Figure 4.1 shows the mean posterior inclusion probabilities of the 28 variables in LR-SS. Among these variables, 4 have inclusion probabilities greater than 0.5, indicating that the model has identified 4 significant variables. Although LR-SS performs better in terms of dimensionality reduction compared to GPR-SS, its predictive performance is not as strong. As shown in Table 4.4, GPR-SS achieves the lowest MSE (0.4772), demonstrating a clear difference from both the standard GPR (0.4937) and LR-SS (0.5021).

LR-SS may overlook the impact of some variables that could significantly influence transfer market value while eliminating most variables. Although the dimensionality

reduction capability of GPR-SS is not as strong as that of LR-SS, it more comprehensively captures the influence of variables on transfer market value. Overall, our proposed GPR-SS model demonstrates superior performance in terms of both effective dimensionality reduction and higher predictive accuracy.

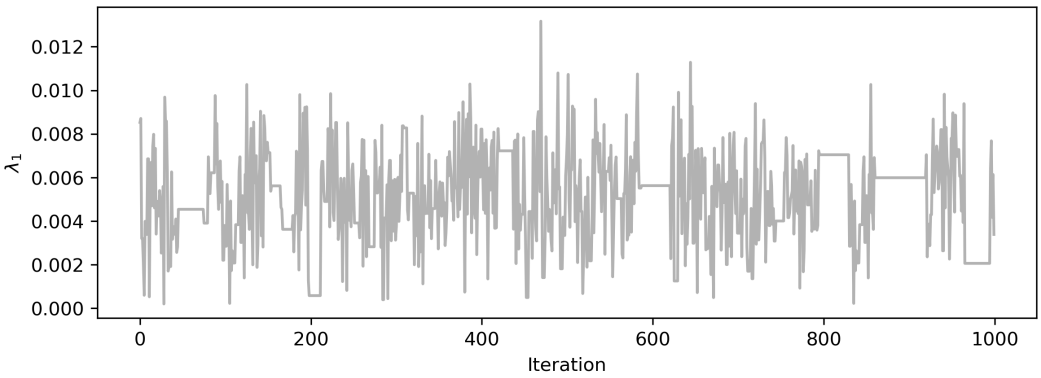


Figure 4.2: *Plot of the First Coordinate in Dimension 28*

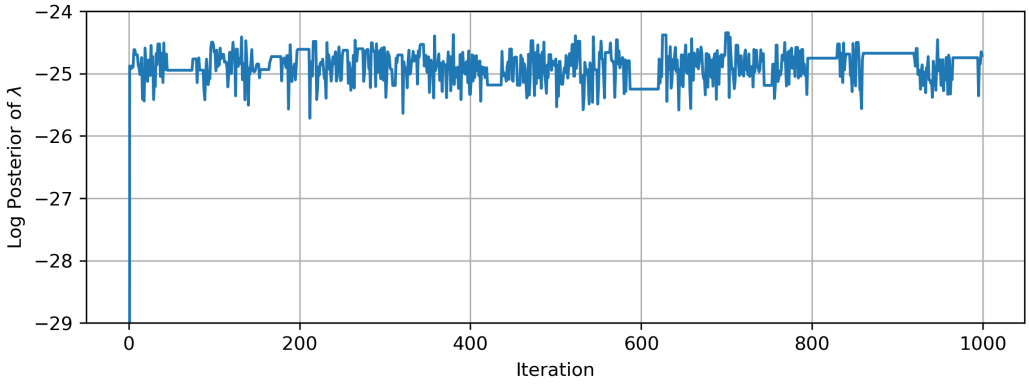


Figure 4.3: *Trace Plot of the Log Posterior*

Figure 4.2 illustrates the trajectory of λ_1 (the inverse length scale of the first variable) during one cross-validation process. The sampling process iterates 1000 times, with an acceptance rate of 56.2%. It can be observed that its value fluctuates within a certain range (other lambdas exhibit similar behaviour, and five more trace plots are shown in the Appendix). Figure 4.3 shows the changes in the logarithm of the posterior distribution of λ, revealing that the values quickly converge. Overall, our sampling method ensures the convergence of the parameters.

Chapter 5

Concluding Remarks

This study introduces a procedure that employs GPR integrated with spike and slab priors for variable selection. This method presents higher predictive accuracy than traditional approaches and has successfully identified 11 significant variables out of 28 football variables, showcasing excellent dimensionality reduction capabilities. The selected 11 variables, such as **goals**, **assists**, and **dribbleWon**, are considered crucial indicators of a forward's performance by common sense, which indicates that GPR-SS accurately identifies significant variables in football data. Furthermore, we can apply GPR-SS to other datasets to discuss its predictive and variable selection performance.

In the sampling process, σ^2 is fixed to accelerate the sampling process. However, it is feasible to assign an informative prior to σ^2, which could potentially enhance the model's robustness and flexibility by integrating domain-specific knowledge or empirical evidence. This adjustment may lead to more accurate posterior distributions and improved inferential performance, but we may need to design a more efficient sampling method.

We assume that the inverse length-scale parameters of each dimension are independent. In future work, we can further investigate the dependencies between these parameters. Understanding these dependencies may lead to more refined models and improved performance, as the interactions among parameters can provide additional insights into the underlying data structure.

We conducted variable selection among 28 variables, but when the dimensionality increases to larger numbers, such as 100 or even 1000, MCMC sampling becomes challenging. As the dimensionality increases, the convergence of MCMC can slow down

significantly, and the chain may require a much larger number of iterations to explore the posterior distribution adequately. Additionally, high-dimensional spaces can cause issues such as multimodality and complex correlation structures among parameters, further complicating the sampling process. To address the challenges of high-dimensional spaces, we can try the Spike and Slab Variational Gaussian Process (SSVGP) (Dance and Paige, 2022), which is a fast and scalable training method in high-dimension and large datasets

Appendix

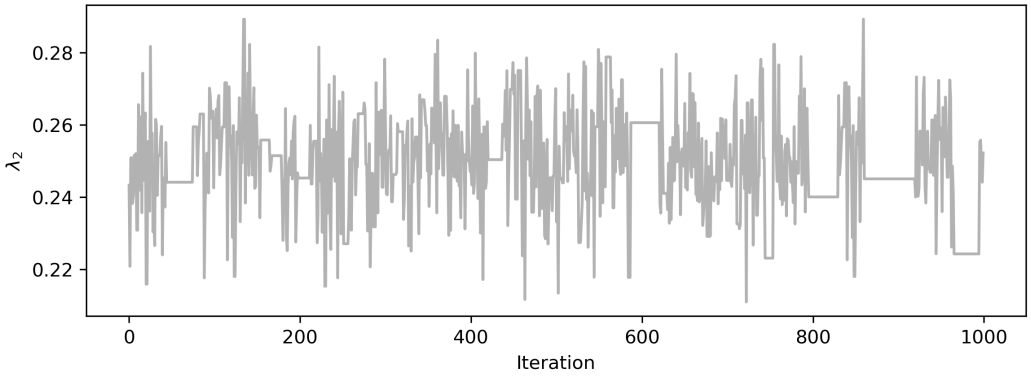


Figure A.1: *Plot of the Second Coordinate in Dimension 28*

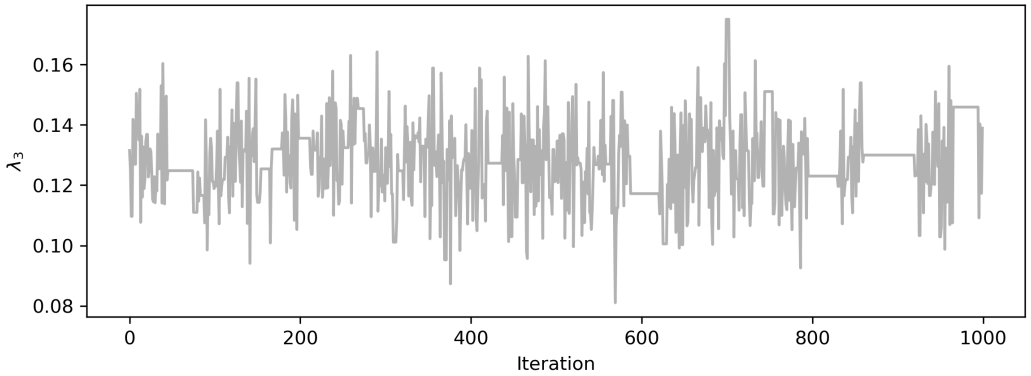


Figure A.2: *Plot of the Third Coordinate in Dimension 28*

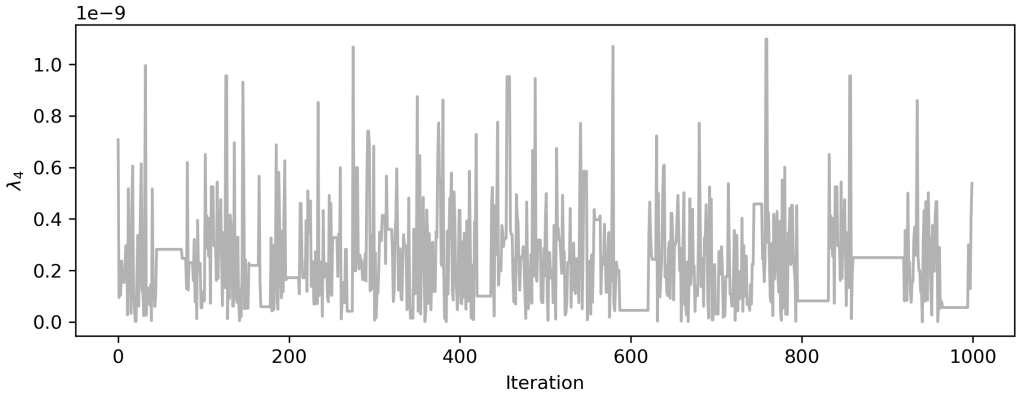


Figure A.3: *Plot of the Fourth Coordinate in Dimension 28*

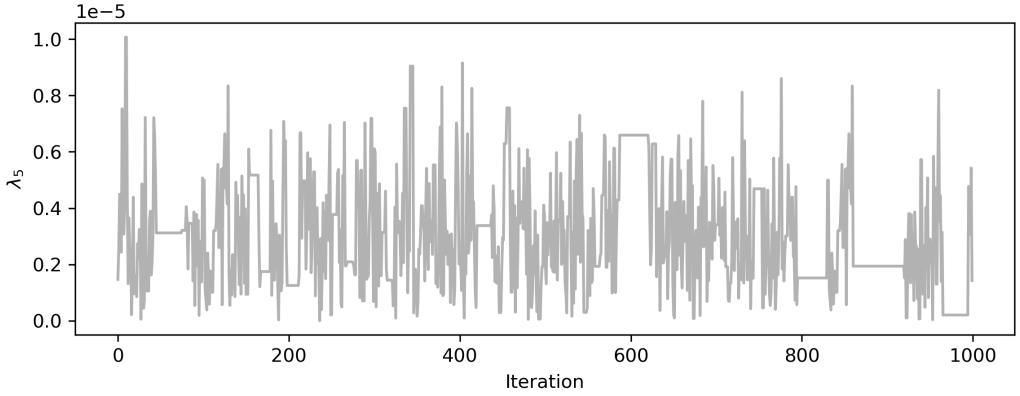


Figure A.4: *Plot of the Fifth Coordinate in Dimension 28*

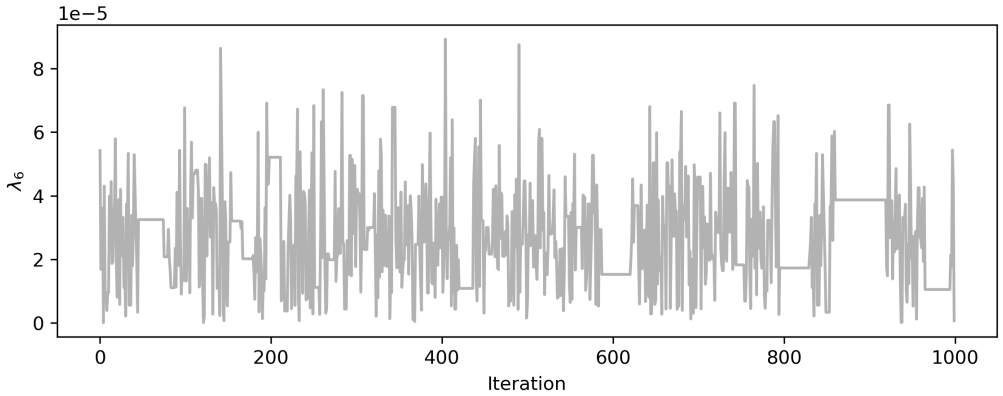


Figure A.5: *Plot of the Sixth Coordinate in Dimension 28*

Bibliography

[1] Andersen, M. R., Winther, O., & Hansen, L. K. (2014). Bayesian inference for structured spike and slab priors. *Advances in Neural Information Processing Systems*, 27.

[2] Casella, G., & George, E. I. (1992). Explaining the Gibbs sampler. *The American Statistician* 46, pp167-174.

[3] Chipman, H., George, E. I., McCulloch, R. E., Clyde, M., Foster, D. P., & Stine, R. A. (2001). The practical implementation of Bayesian model selection. *Lecture Notes-Monograph Series*, pp65-134.

[4] Castillo, I., Schmidt-Hieber, J., & Van der Vaart, A. (2015). Bayesian linear regression with sparse priors. *The Annals of Statistics* 43, pp1986-2018.

[5] Dance, H., & Paige, B. (2022). Fast and scalable spike and slab variable selection in high-dimensional Gaussian processes. In *International Conference on Artificial Intelligence and Statistics*, pp7976-8002. PMLR.

[6] Fonti, V., & Belitser, E. (2017). Feature selection using LASSO. *VU Amsterdam Research Paper in Business Analytics* 30, pp1-25.

[7] Geyer, C. J. (1992). Practical Markov chain Monte Carlo. *Statistical Science*, pp473-483.

[8] George, E. I., & McCulloch, R. E. (1993). Variable selection via Gibbs sampling. *Journal of the American Statistical Association* 88, pp881-889.

[9] George, E. I., & McCulloch, R. E. (1997). Approaches for Bayesian variable selection. *Statistica Sinica* 7, pp339-373.

[10] Hensman, J., Fusi, N., and Lawrence, N. D. (2013). Gaussian processes for big data. In *Proceedings of the Twenty-Ninth Conference on Uncertainty in Artificial Intelligence*, pp282–290.

[11] Ishwaran, H., & Rao, J. S. (2005). Spike and slab variable selection: Frequentist and Bayesian strategies. *The Annals of Statistics* 33, pp730-773.

[12] Linkletter, C., Bingham, D., Hengartner, N., Higdon, D., & Ye, K. Q. (2006). Variable selection for Gaussian process models in computer experiments. *Technometrics* 48, pp478-490.

[13] Miller, A. (2002). *Subset Selection in Regression*. Chapman and Hall/CRC.

[14] MacKay, D. J. et al. (1998). Introduction to Gaussian processes, in C. M. Bishop (ed.), *NATO ASI Series F Computer and Systems Sciences*. Springer.

[15] Mehmood, T., Liland, K. H., Snipen, L., & Sæbø, S. (2012). A review of variable selection methods in partial least squares regression. *Chemometrics and Intelligent Laboratory Systems* 118, pp62-69.

[16] Natarajan, B. K. (1995). Sparse approximate solutions to linear systems. *SIAM Journal on Computing* 24, pp227-234.

[17] Narisetty, N. N., & He, X. (2014). Bayesian variable selection with shrinking and diffusing priors. *The Annals of Statistics* 42, pp789-817.

[18] Pillai, N. S., Stuart, A. M., & Thiéry, A. H. (2011). On the random walk Metropolis algorithm for Gaussian random field priors and gradient flow. *Arxiv preprint*.

[19] Paananen, T., Piironen, J., Andersen, M. R., & Vehtari, A. (2019). Variable selection for Gaussian processes via sensitivity analysis of the posterior predictive distribution. In *The 22nd International Conference on Artificial Intelligence and Statistics*, pp1743-1752. PMLR.

[20] Penny, W., Kiebel, S., & Friston, K. (2006). *Variational Bayes*. Elsevier, London.

[21] Raftery, A. E., Madigan, D., & Hoeting, J. A. (1997). Bayesian model averaging for linear regression models. *Journal of the American Statistical Association* 92, pp179-191.

[22] Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. MIT Press.

[23] Rasmussen, C. E. (2003). Gaussian processes in machine learning. In *Summer School on Machine Learning*, pp63-71. Berlin, Heidelberg: Springer Berlin Heidelberg.

[24] Roberts, G. O., & Rosenthal, J. S. (2009). Examples of adaptive MCMC. *Journal of Computational and Graphical Statistics* 18, pp349-367.

[25] Rue, H., Martino, S., & Chopin, N. (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 71, pp319-392.

[26] Rossi, S., Heinonen, M., Bonilla, E., Shen, Z., & Filippone, M. (2021, March). Sparse Gaussian processes revisited: Bayesian approaches to inducing-variable approximations. In *International Conference on Artificial Intelligence and Statistics*, pp1837-1845. PMLR.

[27] Sánchez-Maroño, N., Alonso-Betanzos, A., & Tombilla-Sanromán, M. (2007). Filter methods for feature selection–a comparative study. In *International Conference on Intelligent Data Engineering and Automated Learning*, pp178-187. Berlin, Heidelberg: Springer Berlin Heidelberg.

[28] Titsias, M., & Lawrence, N. D. (2010). Bayesian Gaussian process latent variable model. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, pp844-851. JMLR Workshop and Conference Proceedings.

[29] Titsias, M., & Lázaro-Gredilla, M. (2011). Spike and slab variational inference for multi-task and multiple kernel learning. *Advances in Neural Information Processing Systems*, 24.

[30] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 58, pp267-288.